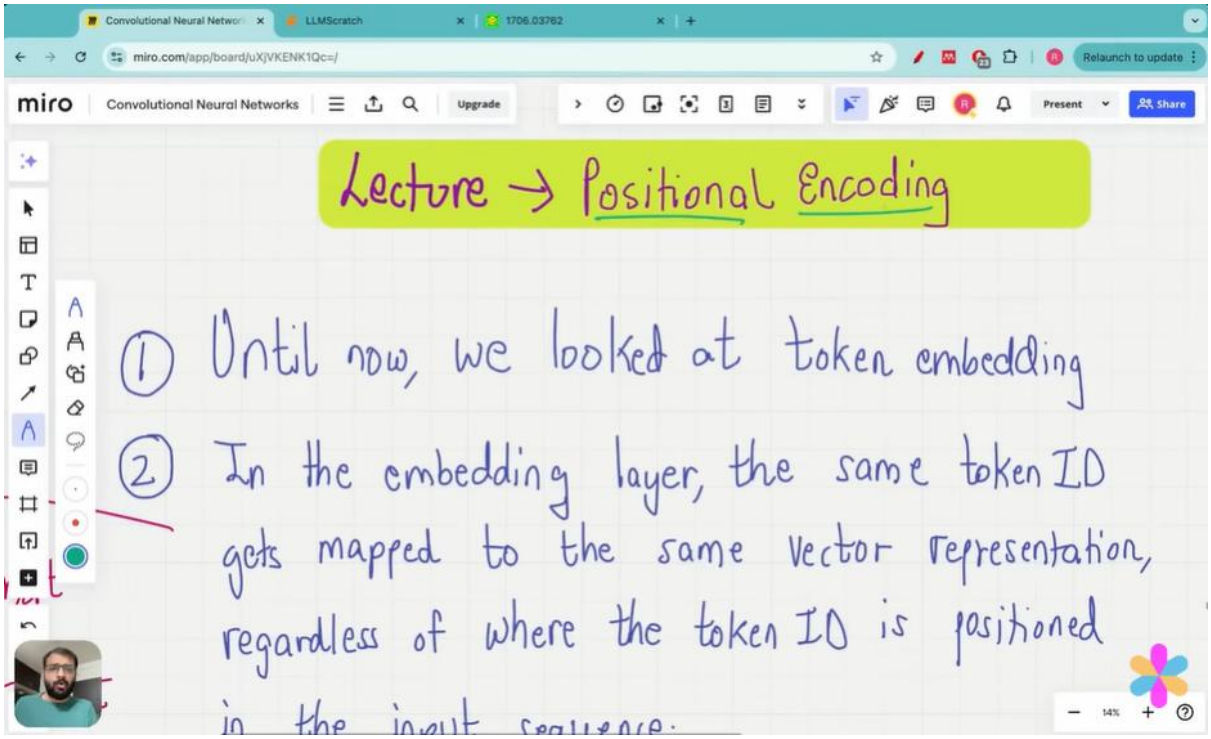
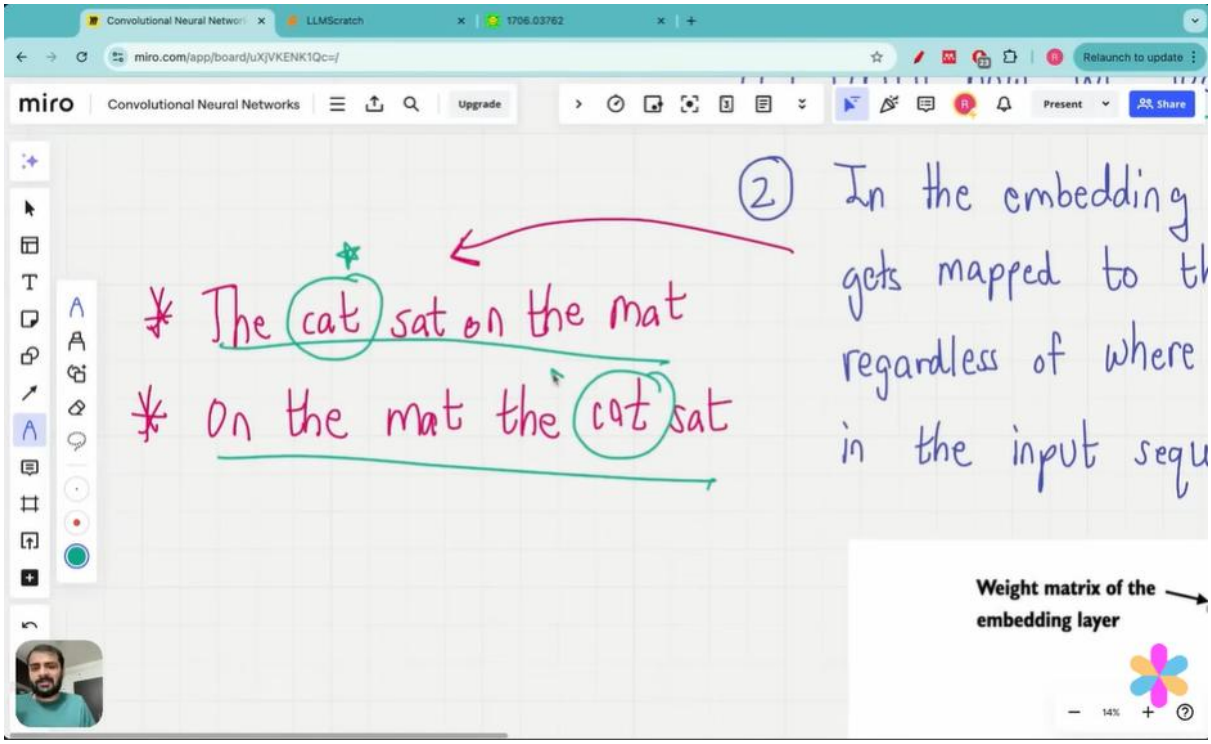
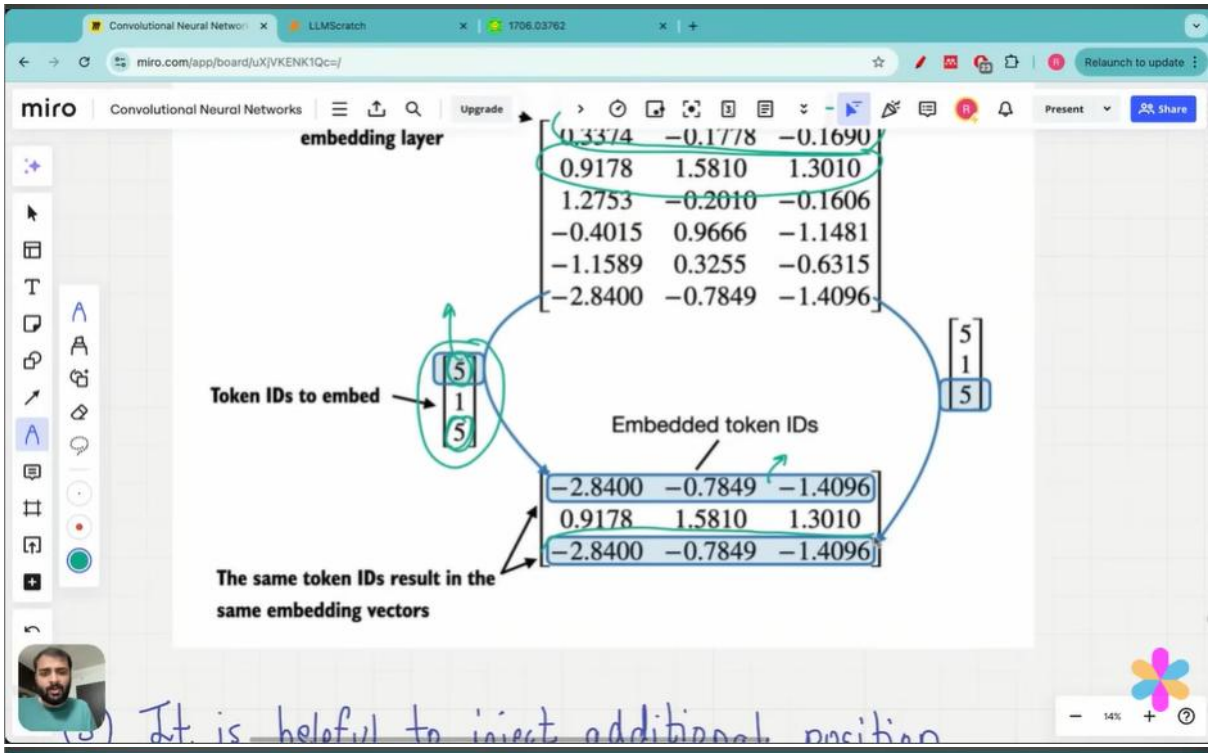
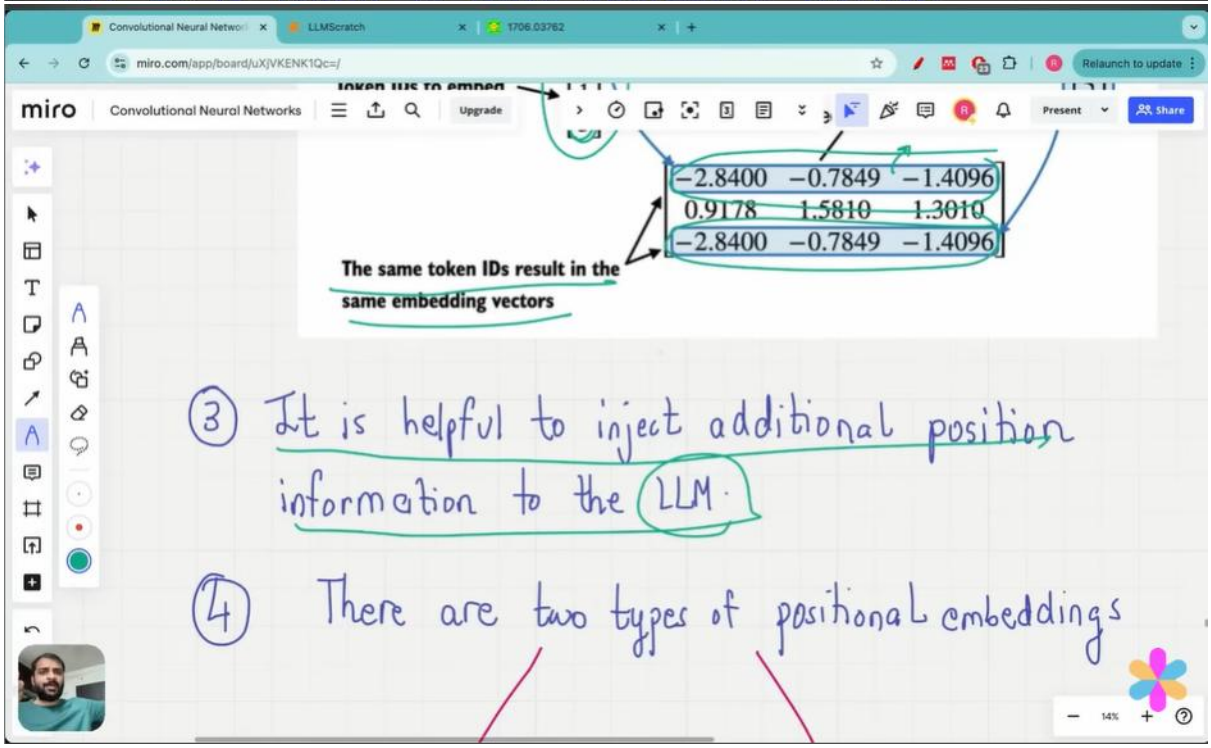
Lec11


Convolutional Neural Networ
LLMScratch
1706.03762
miro.com/app/board/uXjVKENK1Qc=/
Relaunch to update
miro
Convolutional Neural Networks
Upgrade
Present
Share
Lecture → Positional Encoding
① Until now, we looked at token embedding
② In the embedding layer, the same token ID gets mapped to the same vector representation, regardless of where the token ID is positioned in the input sequence.
14%


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Present
Share
② In the embedding
gets mapped to th
regardless of where
in the input sequ
* The cat sat on the mat
* On the mat the cat sat
Weight matrix of the embedding layer
14%

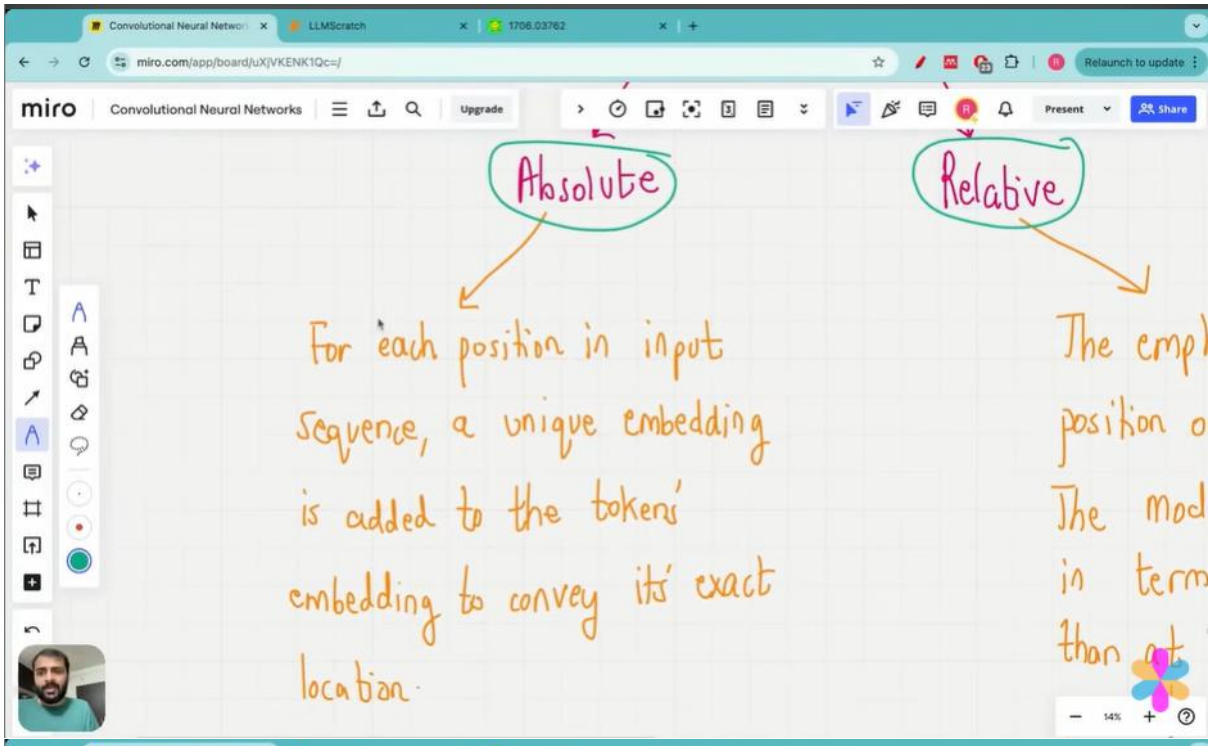


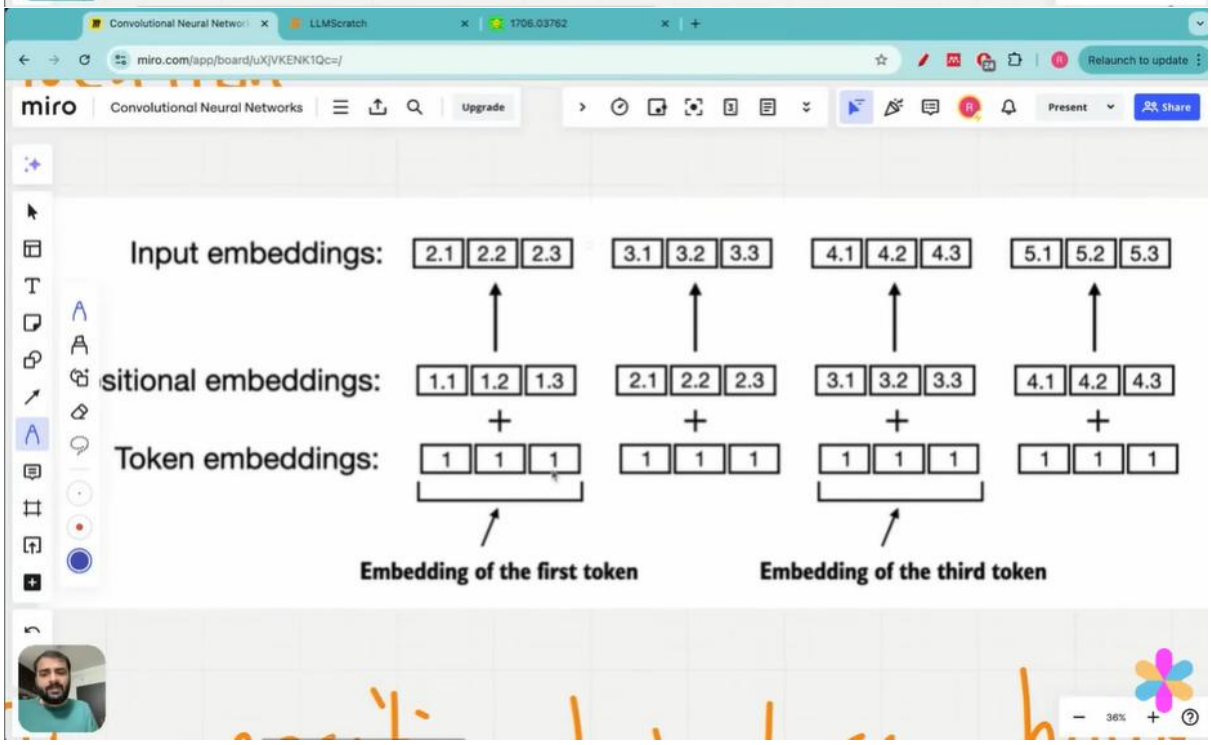
(3) It is helpful to inject additional position

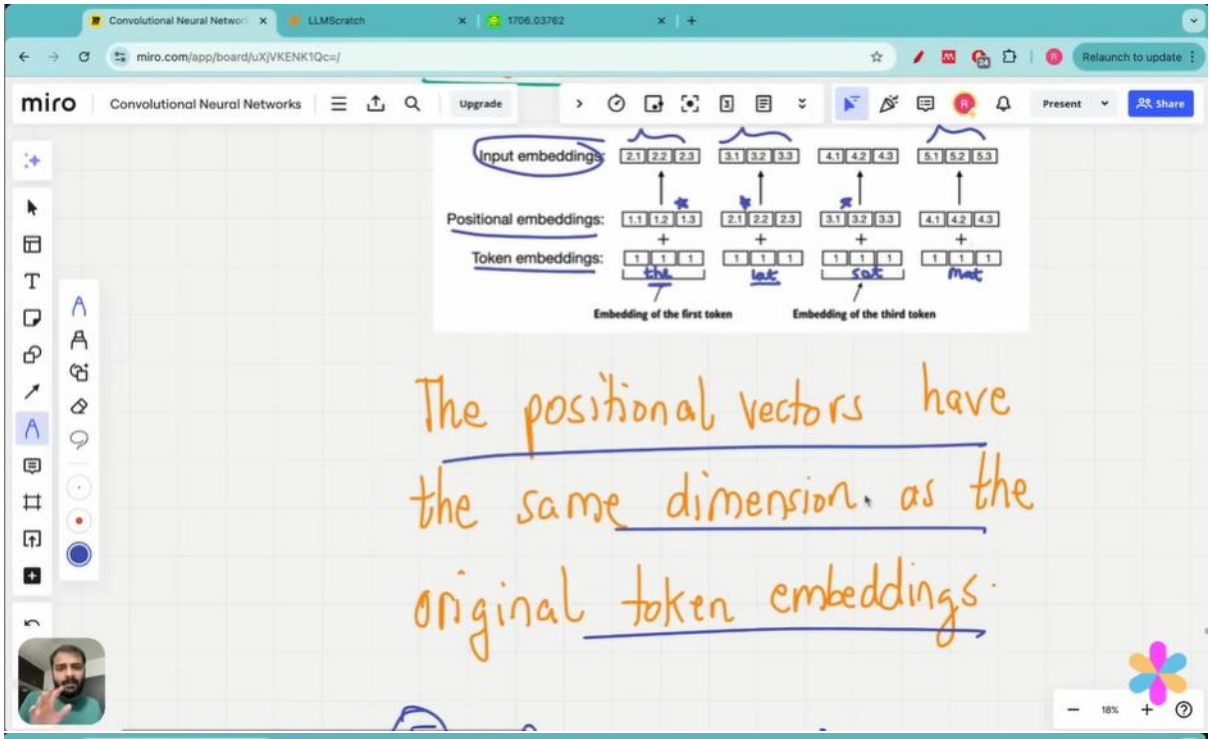


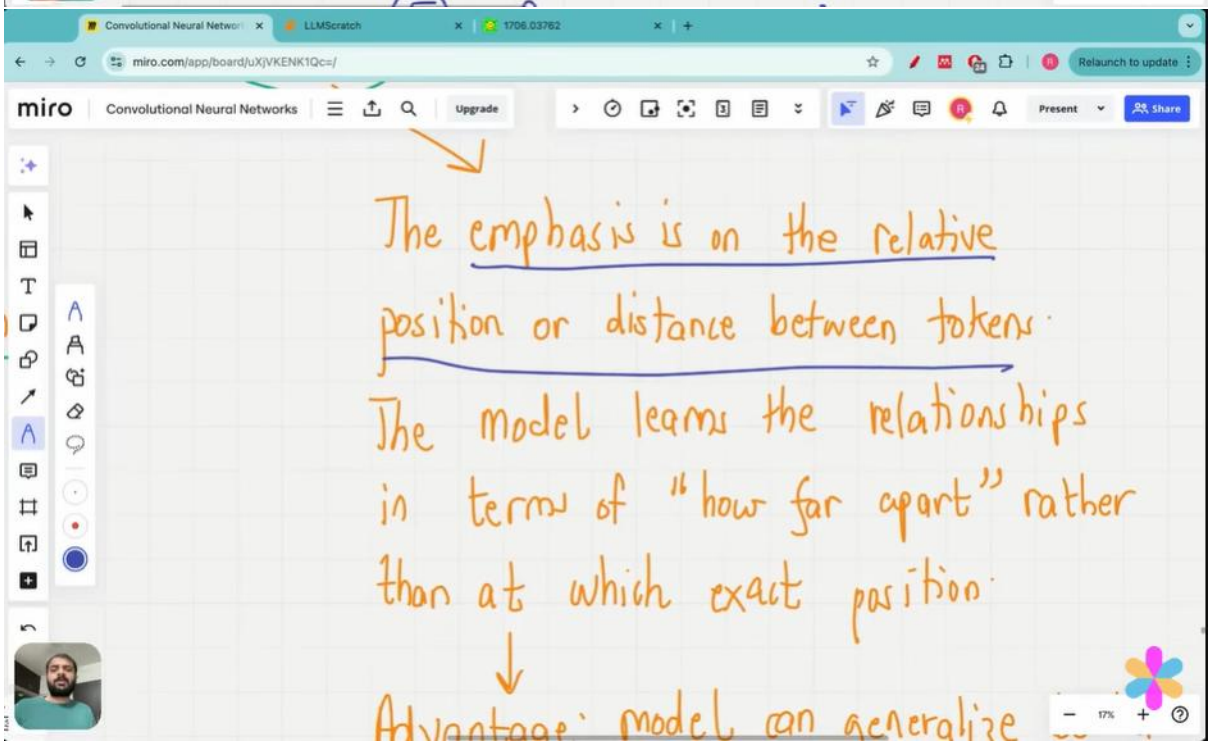
(3) It is helpful to inject additional position information to the LLM.

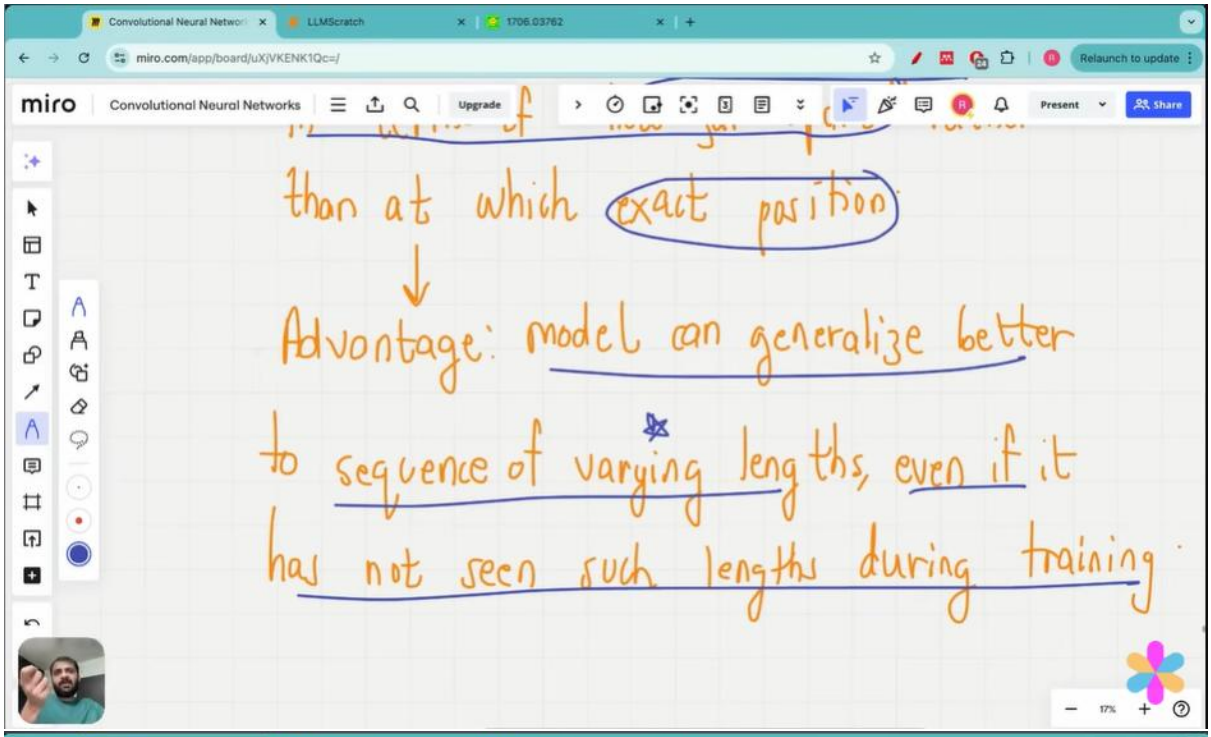
(4) There are two types of positional embeddings

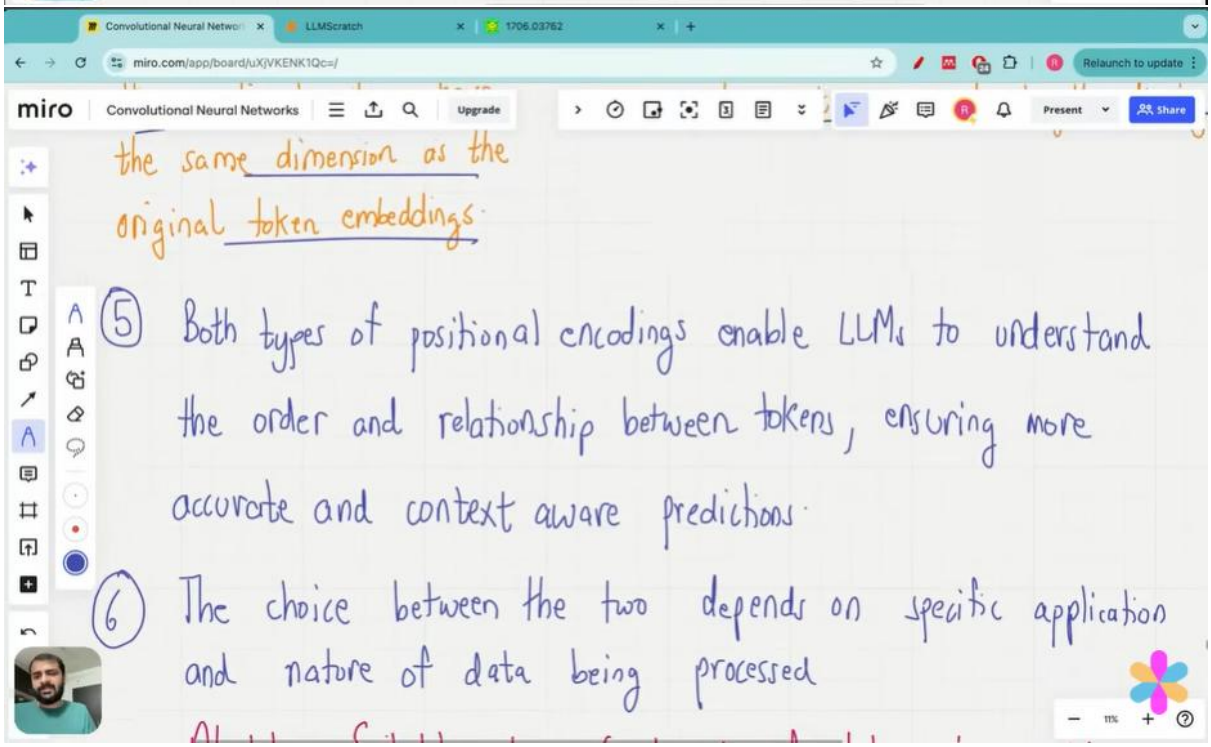

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Present
Share
Absolute
Relative
For each position in input sequence, a unique embedding is added to the tokens' embedding to convey its exact location.
The empl
position o
The mod
in term
than at
14%

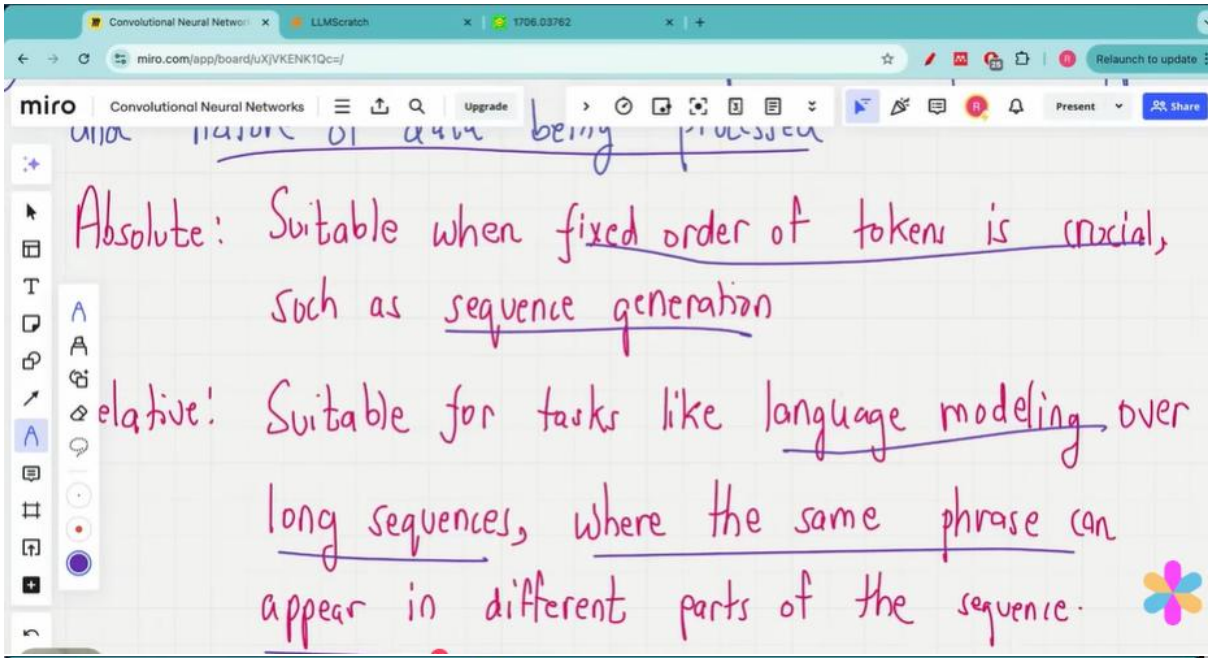

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Present
Share
Input embeddings:
2.1 2.2 2.3
3.1 3.2 3.3
4.1 4.2 4.3
5.1 5.2 5.3
sitional embeddings:
1.1 1.2 1.3
2.1 2.2 2.3
3.1 3.2 3.3
4.1 4.2 4.3
Token embeddings:
1 1 1
Embedding of the first token
Embedding of the third token
36%

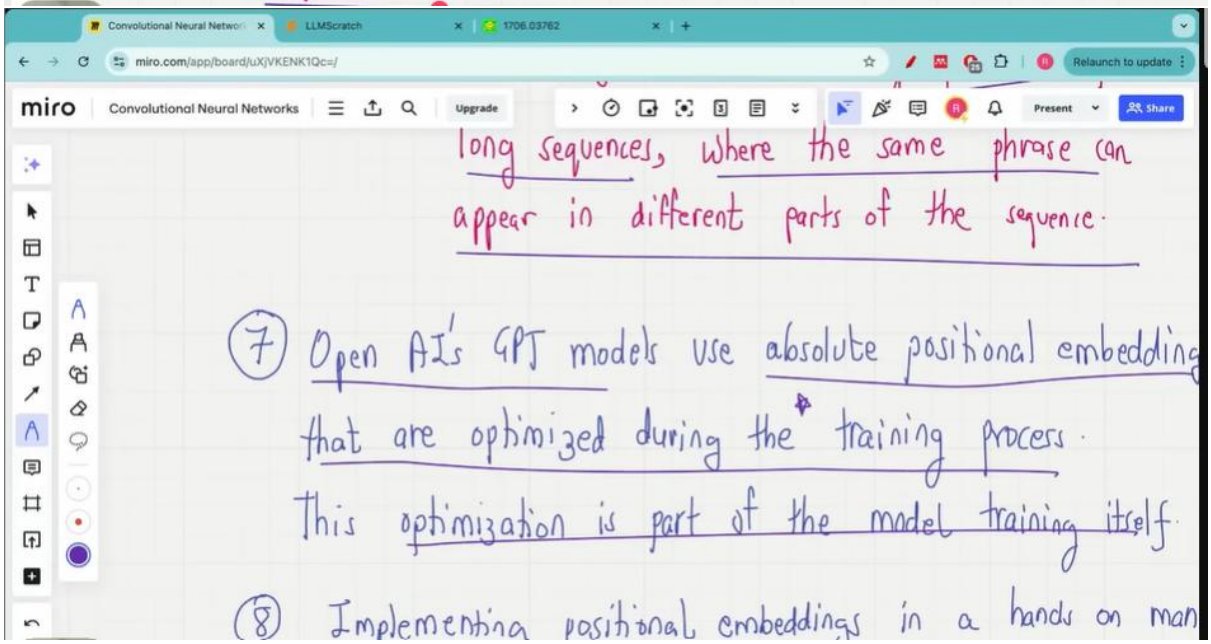

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Present
Share
Input embeddings:
Positional embeddings:
Token embeddings:
Embedding of the first token
Embedding of the third token
The positional vectors have the same dimension as the original token embeddings.
18%

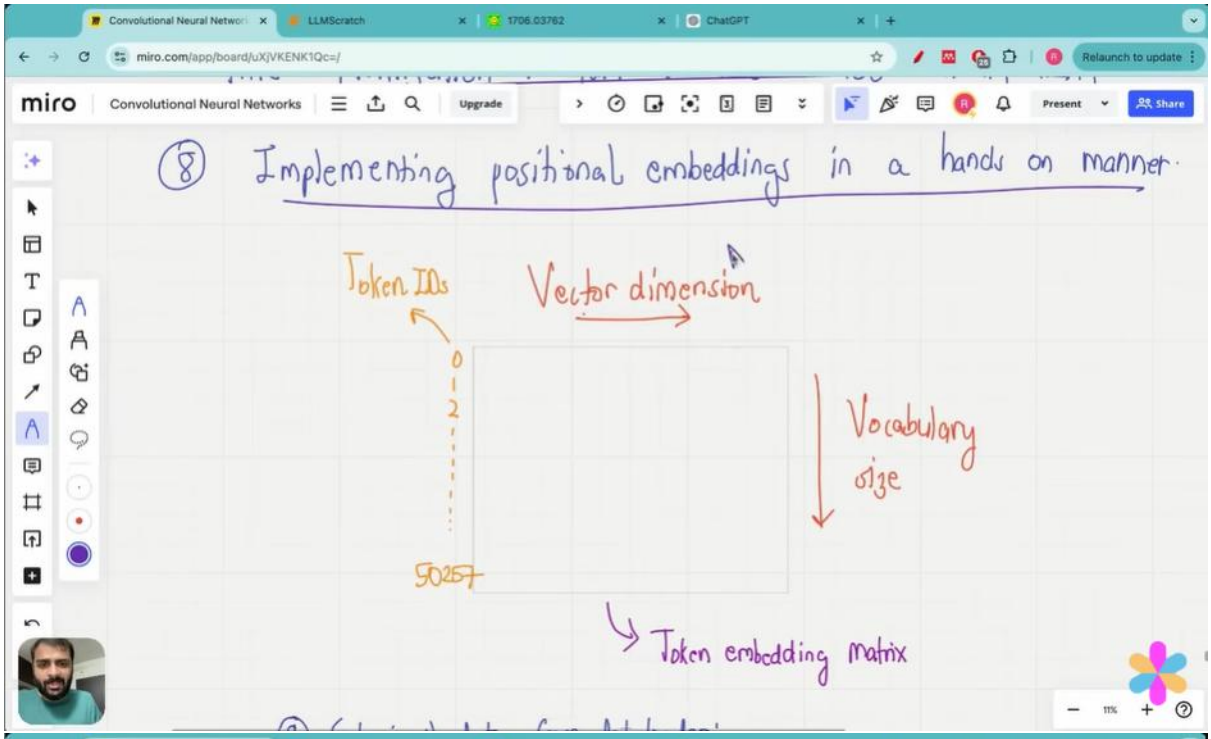

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The emphasis is on the relative position or distance between tokens.
The model learns the relationships in terms of "how far apart" rather than at which exact position.
Advantage: model can generalize
17%

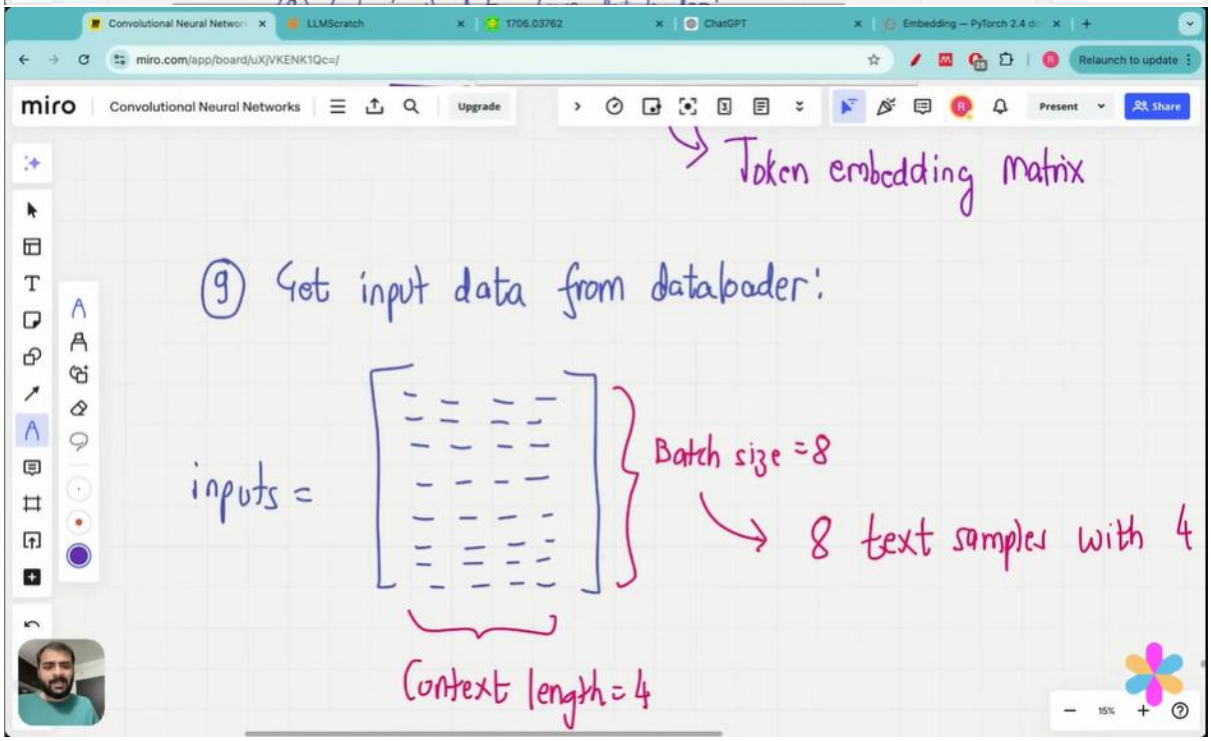

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than at which exact position
Advantage: model can generalize better
to sequence of varying lengths, even if it
has not seen such lengths during training.
17%

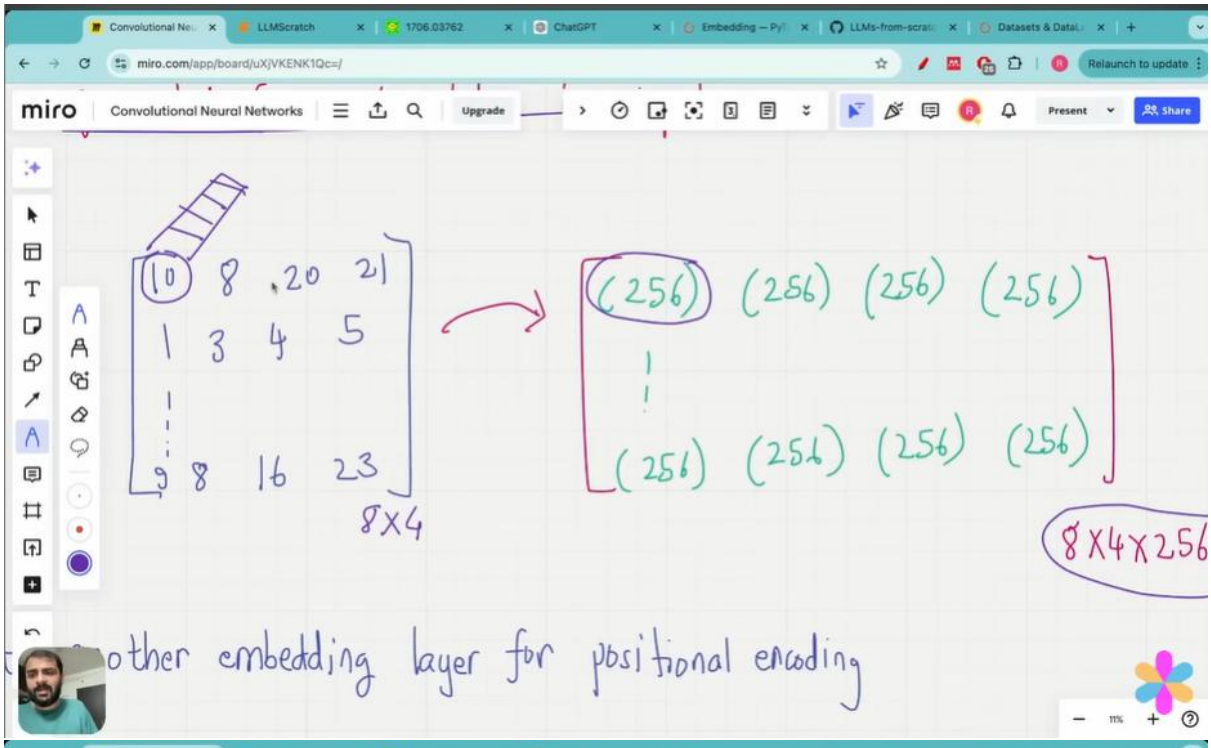

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Relaunch to update
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Convolutional Neural Networks
Upgrade
Present
Share
the same dimension as the
original token embeddings.
(5) Both types of positional encodings enable LLMs to understand the order and relationship between tokens, ensuring more accurate and context aware predictions.
(6) The choice between the two depends on specific application and nature of data being processed
11%

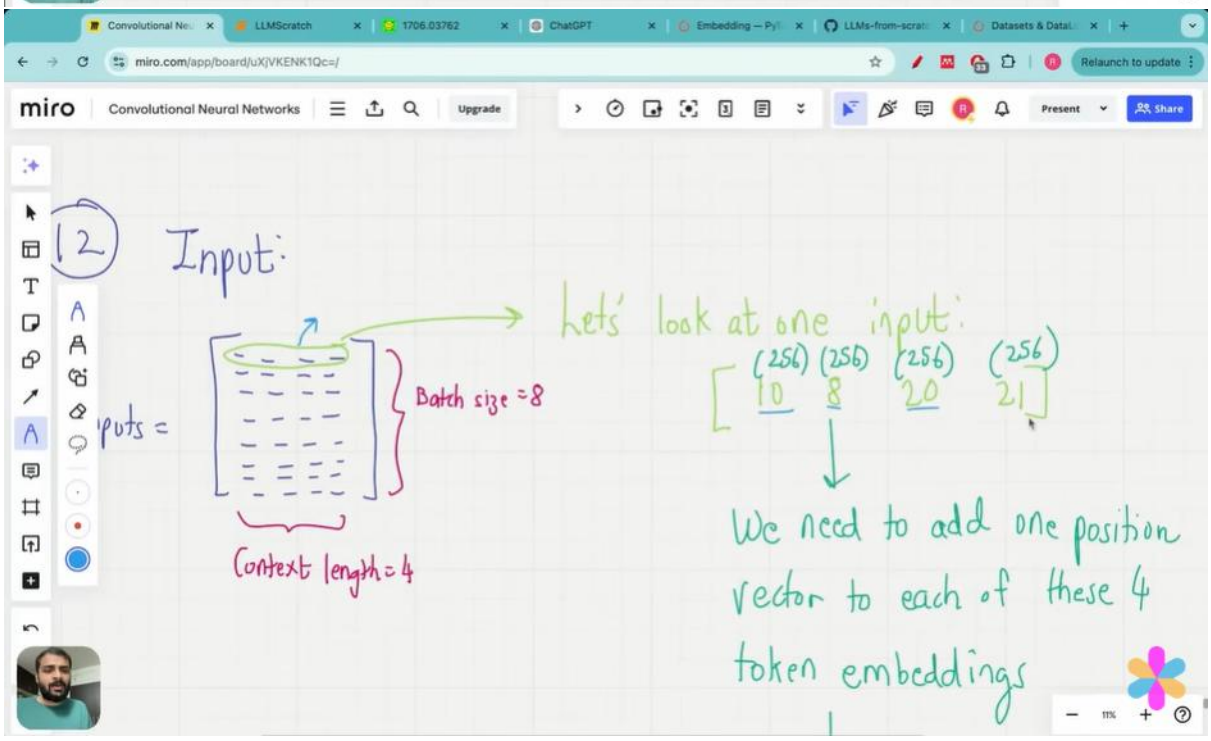

Convolutional Neural Networ
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Convolutional Neural Networks
Upgrade
Present
Share
Absolute: Suitable when fixed order of tokens is crucial, such as sequence generation
elative: Suitable for tasks like language modeling over long sequences, where the same phrase can appear in different parts of the sequence.

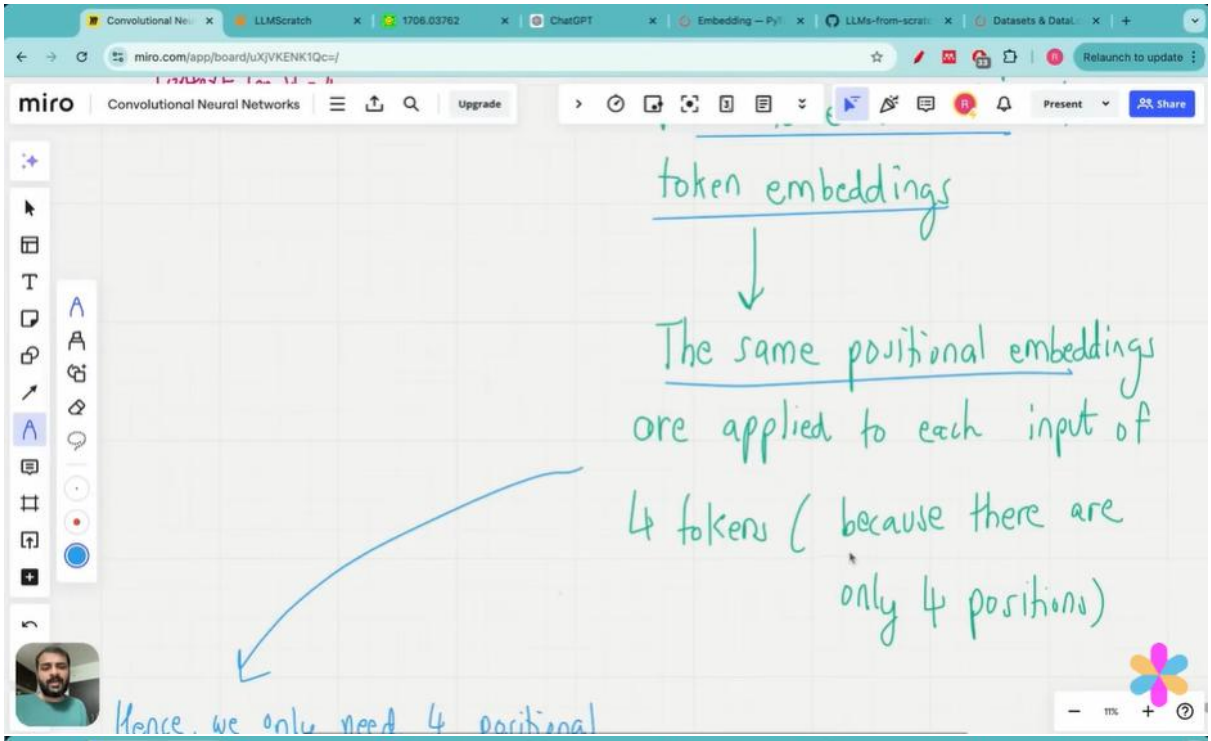

Convolutional Neural Networ
LLMScratch
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miro.com/app/board/uXjVKENK1Qc=/
Relaunch to update
miro
Convolutional Neural Networks
Upgrade
Present
Share
long sequences, where the same phrase can appear in different parts of the sequence.
(7) Open AI's GPT models use absolute positional embedding that are optimized during the training process. This optimization is part of the model training itself.
(8) Implementing positional embeddings in a hands on man

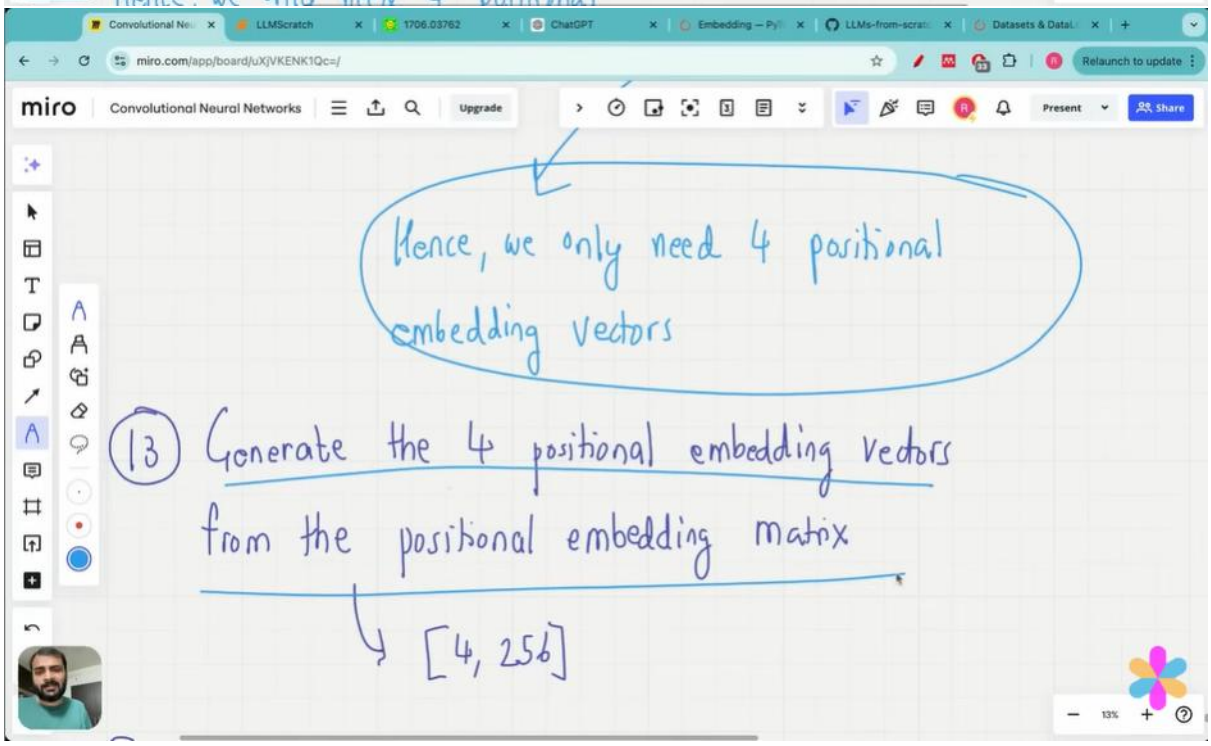

Convolutional Neural Networks
LLMScratch
1706.03762
ChatGPT
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Relaunch to update
miro
Convolutional Neural Networks
Upgrade
Present
Share
⑧ Implementing positional embeddings in a hands on manner.
Token IDs
Vector dimension
0
1
2
50257
Vocabulary size
Token embedding matrix
11%

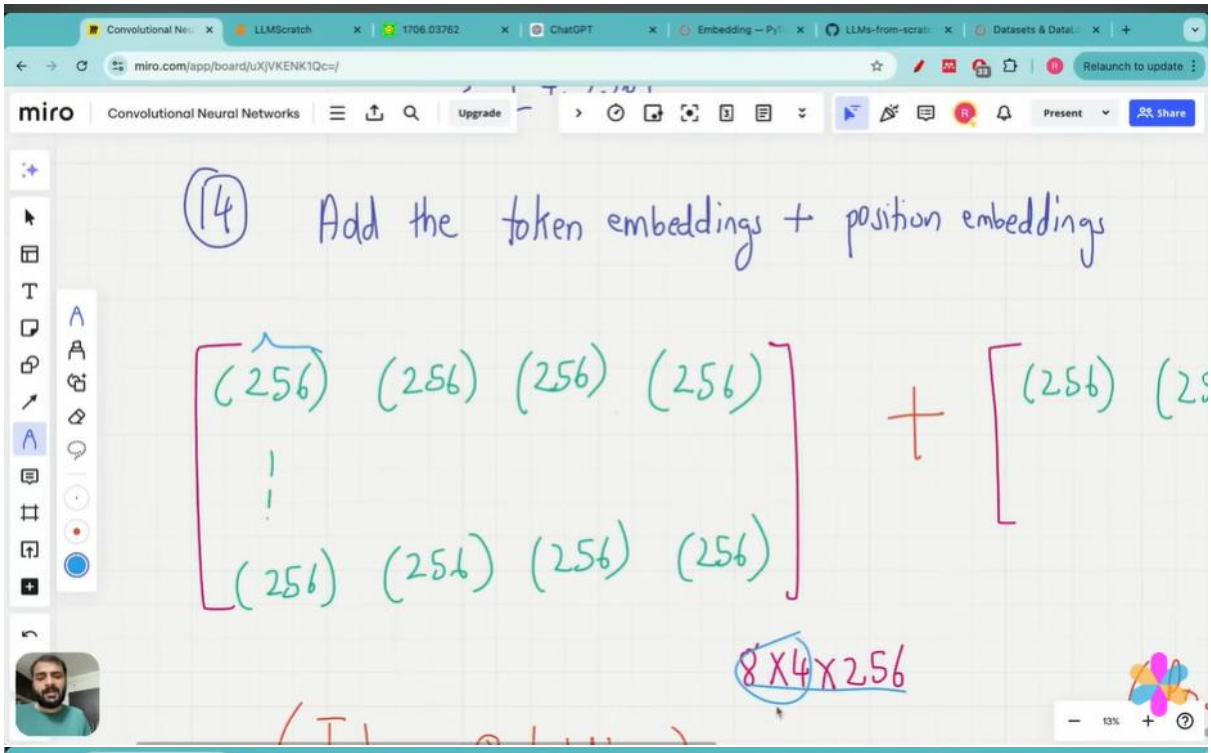

Convolutional Neural Networks
LLMScratch
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ChatGPT
Embedding — PyTorch 2.4
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Convolutional Neural Networks
Upgrade
Present
Share
Token embedding matrix
⑨ Get input data from dataloader:
inputs =
Batch size = 8
8 text samples with 4
Context length = 4
15%

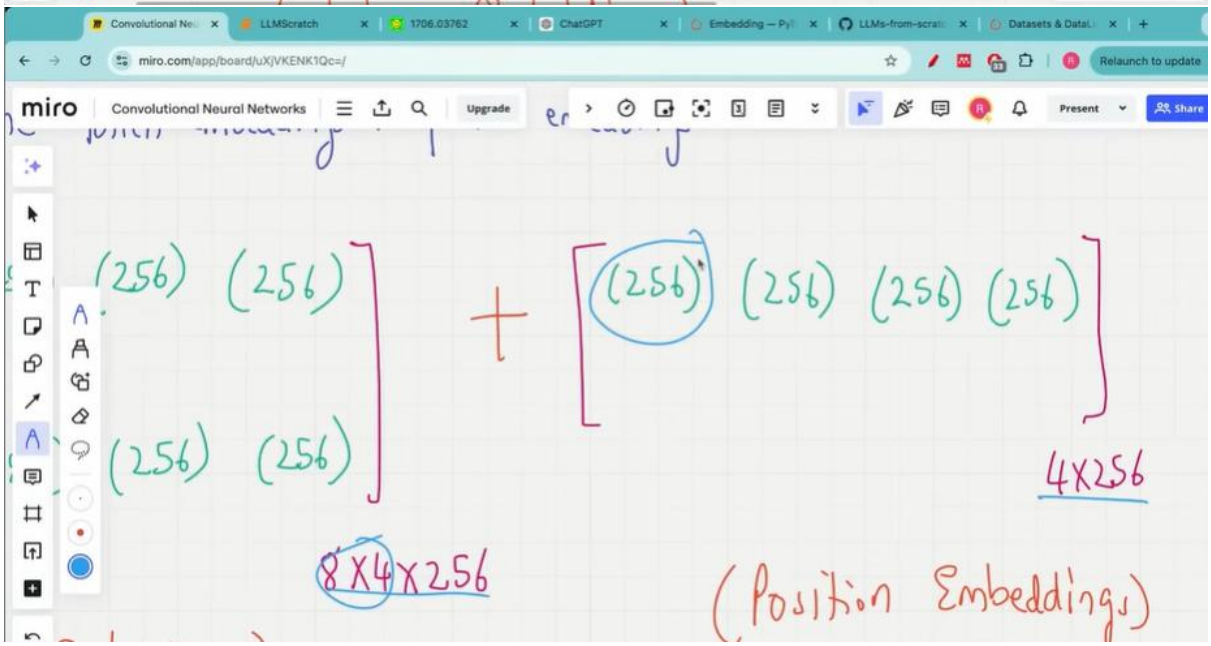

miro
Convolutional Neural Networks
Upgrade
Present
Share
10 8 20 21
1 3 4 5
9 8 16 23
8X4
(256) (256) (256) (256)
(256) (256) (256) (256)
8X4X256
other embedding layer for positional encoding


miro
Convolutional Neural Networks
Upgrade
Present
Share
12
Input:
Batch size = 8
Context length = 4
Let's look at one input:
(256) (256) (256) (256)
10 8 20 21
We need to add one position vector to each of these 4 token embeddings


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Upgrade
Present
Share
token embeddings
The same positional embeddings are applied to each input of 4 tokens (because there are only 4 positions)
Hence, we only need 4 positional


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Convolutional Neural Networks
Upgrade
Present
Share
Hence, we only need 4 positional embedding vectors
13 Generate the 4 positional embedding vectors from the positional embedding matrix
[4, 256]


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Convolutional Neural Networks
Upgrade
Present
Share
(4) Add the token embeddings + position embeddings
(256) (256) (256) (256)
(256) (256) (256) (256)
8X4X256


miro
Convolutional Neural Networks
Upgrade
Present
Share
(256) (256)
(256) (256)
(256) (256) (256) (256)
4X256
8X4X256
(Position Embeddings)